CEHD iPad Initiative: Student Perspectives on Use and Engagement

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Study Background

In fall 2010, the College of Education and Human Development (CEHD) launched a mobile learning pilot project to examine how iPads could be used for teaching and learning. In the project's second year, starting in fall 2011, the focus was to understand in more depth how first-year students used iPads in their formal and informal learning environments. At the beginning of Fall Semester 2011, iPads were distributed to the incoming first-year class. Within their classes, students encountered a range of assignments, from reading and annotating PDFs on iPads, to creating "digital stories" using the iPad's recording capabilities, to field-based projects involving data collection and analysis. Students were also allowed to

Purpose

For this study we looked at how TRiO students responded to the survey items with respect to the iPad and their technology identity, educational experience and daily usage. As more higher education institutions are adopting mobile technology initiatives, this study provides information about one particular population and the CEHD iPad initiative.

Methodology

At the end of Year 2, students who were enrolled in a Learning Community course were asked to complete a survey. Out of 349 learning community student 274 completed the student survey (78.5%). The project team developed a survey that focused on first-year students' perceptions, usage, and educational experiences with respect to the iPad and their learning. Specifically, the survey comprised three main constructs:

- Technology self-concept
- iPad usage in class and out of class activities
- iPad enhancement of the student learning experience

In addition to the three constructs, we asked students to respond to three open-ended questions.

1. Please describe how you use your iPad on a typical day.
2. Please describe the BEST way that the iPad was used in connection with one of your classes.
3. If you could tell your instructors anything about your experience using the iPad for course-related activities, what would you tell them?

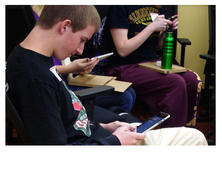
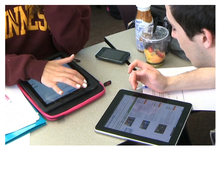
Response Rates

2011-2012 Survey	Total	Non-TRiO	TRiO	Female	Male
Total # of first-year students	433	282	151	288 (92 in TRiO)	145 (59 in TRiO)
Total # of iPads distributed	452				
Students enrolled in learning communities	351	244	127	231	120
Students who took the survey	280				
Complete responses included in analysis	274	180	94	191	83
Response rate (based on complete responses)	78%	80%	74%	83%	69%

Research Question

How did first-year CEHD students respond to questions on their technology identity, their iPad use, and the relationship between the iPad and their educational experience? Did TRiO students' responses differ in any significant way from their non-TRiO counterparts?




Technology Identity

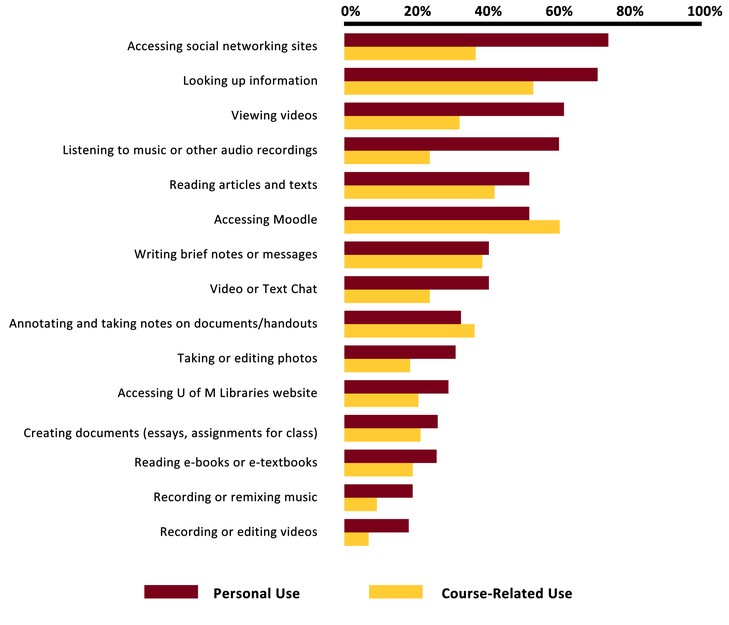
A number of survey questions addressed students' perceptions of themselves and confidence in relation to technology. Students surveyed generally expressed a high degree of confidence in their abilities to use technology and the iPads in their coursework and personal lives. Survey results also show that TRiO students expressed strong confidence in their abilities to use technology/iPads in their coursework--often to an even greater extent than non-TRiO students.

TRiO students were also significantly more likely to engage in information seeking and sharing around technology products than their non-TRiO peers. TRiO students were significantly more likely to describe themselves as liking to read about advances in technology, frequently looking for new software or apps, and discussing technology products with friends.

	TRiO (n=94)				Non-TRiO (n=180)				Sig.	χ^2	ϕ_c
	SD	D	A	SA	SD	D	A	SA			
a. My friends would describe me as "into"the latest technology	5.7(5)	25(22)	44(39)	25(22)	8.2(14)	32.2(55)	42.1(72)	17.5(30)		3.175	0.111
b. I am enthusiastic about consumer electronics and digital devices	4.5(4)	28.2(18)	46.1(41)	29.2(26)	4.1(7)	8.3(14)	56.2(95)	31.4(53)	0.01	10.741	0.203
c. I am excited about the internet	0(0)	10.1(9)	47.2(42)	42.7(38)	4.1(7)	8.3(14)	56.2(95)	31.4(53)	0.075	6.923	0.164
d. I like to read about advances in consumer electronics and digital services	12.4(11)	39.3(35)	34.8(31)	13.5 (12)	31.8(54)	48.8(83)	12.9(22)	6.5(11)	0.001	26.836	0.322
e. I frequently look for new software or apps	10.1(9)	34.8(31)	31.5(29)	23.6(21)	22.8(39)	39.8(68)	29.8(51)	7.6(13)	0.001	16.985	0.256
f. I often tell friends about technology products that interest me	8.0(7)	35.2(31)	37.5(33)	19.3(17)	16.4(28)	46.2(79)	31.0(53)	6.4(11)	0.002	14.358	0.235

iPad Use

Percent of TRiO students who reported using their iPad "at least once a day" for...



Educational Experience

Students generally responded very positively to a number of questions on how the iPad engaged them in their learning experience. A majority of students agreed that the iPads helped engage them in the learning process. A majority of students also reported that they believed receiving an iPad was important for their learning.

TRiO students, however, were significantly more likely than their non-TRiO peers to report that the iPads promoted engagement in their learning experience--a relationship which was robust and highly significant across a number of survey items. Some of the benefits were organizational: TRiO students were significantly more likely to report that the iPads were a helpful time management tool. Other benefits were interpersonal: TRiO students, for example, reported that the iPads promoted class discussion and helped them develop connections with their instructors to a much greater extent than did non-TRiO students.

	TRiO				Non-TRiO				Sig.	χ^2	ϕ_c
	SD	D	A	SA	SD	D	A	SA			
a. Engaged me in the learning process	1.1(1)	14.6(13)	46.1(41)	38.2(34)	12.3(21)	24.6(42)	47.4(81)	15.8(27)	0.0001	23.907	0.303
b. Facilitated multiple types of learning activities	2.2(2)	5.6(5)	55.1(49)	37.1(33)	8.8(15)	15.9(27)	54.7(93)	20.6(35)	0.002	14.882	0.240
c. Helped me develop confidence in my understanding of course material	4.5(4)	23.6(21)	41.6(37)	30.3(27)	13.5(23)	34.5(59)	40.4(69)	11.7(20)	0.0001	18.058	0.264
d. Promoted discussion in class	7.9(7)	23.6(21)	43.8(39)	24.7(22)	12.9(22)	36.8(63)	40.4(69)	9.9(17)	0.01	13.182	0.225
e. Helped me develop connections with my instructor	5.6(5)	21.3(19)	42.7(38)	30.3(27)	14.1(24)	41.8(71)	36.5(62)	7.6(13)	0.0001	30.837	0.345
f. Helped me manage my time	11.2(10)	22.5(20)	40.4(36)	25.8(23)	19.9(34)	42.7(73)	25.1(43)	12.3(21)	0.000	20.149	0.278
g. I believe receiving an iPad was important for my learning	6.7(6)	15.7(14)	41.6(37)	36(32)	18.7(32)	25.1(43)	37.4(64)	18.7(32)	0.01	15.435	0.244

Quotes from Students



"I use my IPad almost everyday when I need to look up assignments on moodle or have access to articles professors give us. On a regular basis, I would always use it for email, taking connects with friends, setting up accounts, and making videos"

"Ipad is a very handy product. I have some class that we need a device that can access to the internet. I always have my Ipad since it is light and easy to carry around. I carried my laptop a couple of time and it was very heavy along with my

"I use it to see what assignments I need to do for classes. I also have used to look up sources for papers. I use it to go on social networking sites, as well to text friends from a text free app."

"I normally use my Ipad and bring it to class when I do not want to bring my laptop or I do not have to use word to write a paper etc. When I do use my Ipad it is normally to check my email, go on moodle and check assignments, as well as..."

"It (iPad) was used to make many videos for my classes. Having the Ipad made it possible to download apps that were helpful for the course."

"Moodle, I was always connected to it and could post thoughts or ideas without worrying about when or where I was located"

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